

Lucy Coe

✉ lucycoe07@gmail.com |  GitHub |  LinkedIn |  Portfolio | StudyFlow

Education

BSc Computer Science, University of Edinburgh

2024–2028

Predicted First Class Honours

Relevant Modules: Algorithms & Data Structures, Computer Systems, Software Engineering, Reasoning & Agents.

Coursework: Cache Simulator (C/Assembly), Student Event Booker (Java)

Technical Skills

Languages: Java, C, Python, JavaScript, SQL
Frameworks: React, Node.js, Tailwind CSS
Backend: REST APIs, Supabase Authentication, Row Level Security
Databases: PostgreSQL, SQL
Testing: Vitest, React Testing Library, JUnit
Tools: Git, Linux (WSL), Vercel

Projects

StudyFlow

React, Supabase, PostgreSQL, Tailwind CSS, Vitest, Vercel

- Designed and built a full-stack study management platform using React, Supabase and PostgreSQL.
- Engineered a constraint-based scheduling algorithm that automatically generates conflict-free study plans around lectures, deadlines and existing calendar events.
- Designed a secure relational database using Supabase Authentication and Row Level Security to isolate user data.
- Built automated tests for scheduling logic using Vitest and React Testing Library.
- Managed development with Git and deployed on Vercel.

Cache Simulator

C, Assembly

- Implemented a cache simulator in C and Assembly to model memory hierarchies, cache replacement behaviour and low-level system performance.

Visual Ciphers

Python, Tkinter

- Built an interactive Python application that visualises classical encryption algorithms through step-by-step execution to aid learning and debugging.

React Mini Projects Collection

React, JavaScript, Tailwind CSS

- Developed 10+ React applications exploring state management, asynchronous programming, REST API integration and reusable component architecture.

Experience

Object-Oriented Programming Tutor and Marker

2026

University of Edinburgh

- Guided 30+ undergraduate students through Java, object-oriented design, debugging and algorithmic problem solving during weekly tutorials.
- Evaluated 15+ coursework submissions and provided structured technical feedback to improve software design, code quality and problem-solving.

Cognitive Science Python Lab Demonstrator

2026

University of Edinburgh

- Supported Python laboratory sessions by diagnosing programming errors, explaining debugging techniques and helping students develop problem-solving skills.

Customer Service and Hospitality Assistant

Until 2024

Café, Restaurant and Pottery Studio

- Delivered customer service in fast-paced hospitality environments, strengthening communication, teamwork and adaptability.

Awards and Achievements

Amazon Future Engineer Scholarship • Robertson Trust Scholarship • Proxime Accessit (Wick High School)